

- 8 (a) Outline the roles of sensory receptor cells in the mammalian nervous system.

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- (b) Fig. 8.1 shows the changes in potential difference (p.d.) across the membrane of a receptor cell over a period of time. The membrane was stimulated at time **A** and at time **B** with stimuli of different intensities.

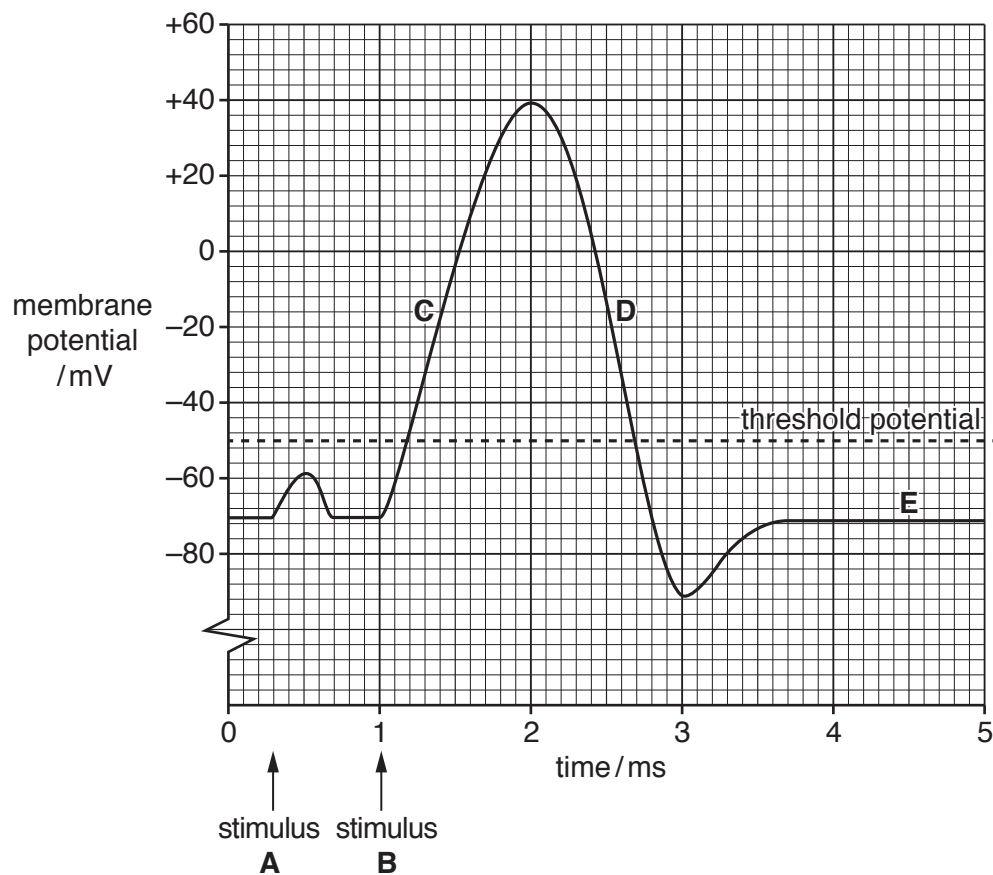


Fig. 8.1

- (i) State which of the letters **C**, **D** and **E** on Fig. 8.1 correspond to each of these events. You may use each of the letters **C**, **D** or **E** once, more than once or not at all.

The  $\text{Na}^+/\text{K}^+$  pump is operating .....

The voltage-gated  $\text{Na}^+$  channels are open .....

The voltage-gated  $\text{K}^+$  channels are open .....

[3]

- (ii) Explain why stimulus **A** did not result in an action potential being produced whereas stimulus **B** did.

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- (iii) Describe the importance of the refractory period in the transmission of action potentials.

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- (iv) Describe how action potentials are transmitted along a myelinated axon.

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[Total: 14]

## Section B

Answer **one** question.